

Task on CloudFront & Route 53

1) Configure VPC peering in cross regions.

—> created VPC #1 in Stockholm (eu-north-1) with CIDR 10.0.0.0/16.
Now let's continue the **cross-region VPC peering setup step-by-step**

The screenshot displays the AWS Management Console for a VPC named 'myvpc-stockholm' (ID: vpc-0acfd10baef05bbe9) in the eu-north-1 region. The 'Details' tab is active, showing various configuration options. The VPC is in an 'Available' state. Key settings include: DNS resolution enabled, Main network ACL (acl-0f8eb0c64dc4dc880), IPv4 CIDR (10.0.0.0/16), and Block Public Access disabled. The 'Resource map' section provides a visual overview of the VPC's components: one subnet (mypublic-sub), two route tables (rtb-0abf2e0a5ea8cc469 and pub-stockholmrt), and one network connection (my-vpcstockigw).

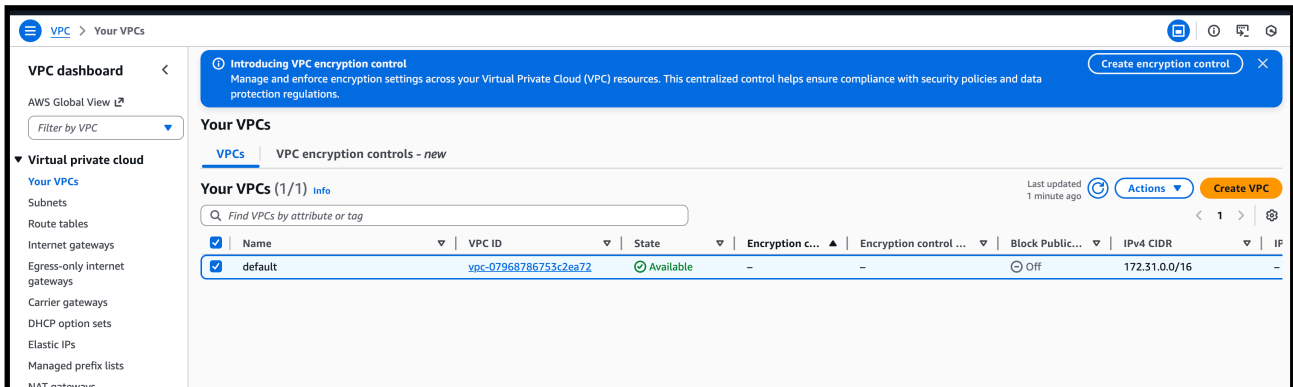
Category	Item	Value
VPC ID	VPC ID	vpc-0acfd10baef05bbe9
State	State	Available
DNS resolution	DNS resolution	Enabled
Main network ACL	Main network ACL	acl-0f8eb0c64dc4dc880
IPv4 CIDR	IPv4 CIDR (Network border group)	10.0.0.0/16
Encryption control ID	Encryption control ID	-
Tenancy	Tenancy	default
Default VPC	Default VPC	No
Network Address Usage metrics	Network Address Usage metrics	Disabled
Encryption control mode	Encryption control mode	-
Block Public Access	Block Public Access	Off
DHCP option set	DHCP option set	dopt-0a1f1caaa13978856
IPv4 CIDR	IPv4 CIDR	10.0.0.0/16
Route 53 Resolver DNS Firewall rule groups	Route 53 Resolver DNS Firewall rule groups	-
DNS hostnames	DNS hostnames	Disabled
Main route table	Main route table	rtb-0abf2e0a5ea8cc469
IPv6 pool	IPv6 pool	-
Owner ID	Owner ID	207662791773

The screenshot shows the 'Your VPCs' page in the AWS Management Console. It lists two VPCs: vpc-01126d967b1bb73af and myvpc-stockholm (vpc-0acfd10baef05bbe9). The 'myvpc-stockholm' VPC is selected, and its details are shown in the table below.

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IP
-	vpc-01126d967b1bb73af	Available	-	-	Off	172.31.0.0/16	-
myvpc-stockholm	vpc-0acfd10baef05bbe9	Available	-	-	Off	10.0.0.0/16	-

—>create a public subnet for this vpc and attach igw to it withsame vpc

- > from another region we have default vpc from north virginia
- > cidr 172.31.0.0/16



- > now go to peering connection and add the details of present vpc region and another vpc details (north virginia)

Navigation: VPC > Peering connections > Create peering connection

Create a tag with a key of 'Name' and a value that you specify.

mystockholm-virginiapeer

Select a local VPC to peer with

VPC ID (Requester)

vpc-0acfd10baef05bbe9 (myvpc-stockholm)

VPC CIDRs for vpc-0acfd10baef05bbe9 (myvpc-stockholm)

CIDR	Status	Status reason
10.0.0.0/16	Associated	-

Select another VPC to peer with

Account

☒ My account

☐ Another account

Region

☐ This Region (eu-north-1)

☒ Another Region

United States (N. Virginia) (us-east-1)

VPC ID (Acceptor)

vpc-07968786753c2ea72

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Value - optional

Q Name X Q mystockholm-virginiapeer X Remove

Add new tag

You can add 40 more tags

- > and click create

—>now go to north virginia region —>vpc —> peering connection—> accept the peering request

The screenshot shows the AWS VPC console's 'Peering connections' page. A green banner at the top states: 'Your VPC peering connection (pcx-0c45278d1f5dfdd8c) has been established. To send and receive traffic across this VPC peering connection, you must add a route to the peered VPC in one or more of your VPC route tables.' Below this, a table lists the peering connections. One connection is shown with a status of 'Provisioning'.

Name	Peering connection ID	Status	Requester VPC	Accepter VPC	Requester CIDRs
-	pcx-0c45278d1f5dfdd8c	Provisioning	vpc-0acfd10baef05bbe9	vpc-07968786753c2ea72 / def...	10.0.0.0/16

This screenshot shows the same AWS VPC console page, but the peering connection status is now 'Active'. The green banner remains. The table below shows the connection is active.

Name	Peering connection ID	Status	Requester VPC	Accepter VPC	Requester CIDRs	Ac
-	pcx-0c45278d1f5dfdd8c	Active	vpc-0acfd10baef05bbe9	vpc-07968786753c2ea72 / def...	10.0.0.0/16	17

—> after accepting add the other vpc cidr range in present virgina vpc route table with peering connection
—> it will remote map to the peering

The screenshot shows the 'Edit routes' page in the AWS VPC console. It displays a table of routes for the route table 'rtb-0b5d29d80613137e7'. The routes include a local route, an Internet Gateway route, and a Peering Connection route. The Peering Connection route is highlighted with a blue selection bar.

Destination	Target	Status	Propagated	Route Origin
172.31.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute
10.0.0.0/16	Peering Connection	-	No	CreateRoute

—> now go inside the Stockholm vpc and add virginia vpc 's cidr range

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute
172.31.0.0/16	Peering Connection	-	No	CreateRoute

[Add route](#) [Cancel](#) [Preview](#) [Save changes](#)

with peering connection

—> now launch the one instance with that vpc from both regions

Instances (1/1)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv
myvirginia-pe...	i-026e730dea2b1c628	Running	t3.micro	Initializing	View alarms +	us-east-1a	ec2-44-202-76-137.co...	44.202.76

Instances (1/3)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv
myec2	i-07ac8ddc64446c905	Stopped	t3.micro	-	View alarms +	eu-north-1a	-	-
mystockpeerin...	i-0bc9dcee84b8c0905	Running	t3.micro	Initializing	View alarms +	eu-north-1c	-	16.171.53.5
myubuntu	i-0115ad88e230a17a3	Stopped	t3.micro	-	View alarms +	eu-north-1b	-	-

—> same in stockholm region also launch ec2

EC2

>

Security Groups

>

sg-08576e6c689924492 - launch-wizard-28

>

Edit inbound rules

🔍

🔍

🔍

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Security group rule ID

sgr-034a96c5a151fb46d

Type [Info](#)

SSH

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source [Info](#)

Custom

Description - optional [Info](#)

Delete

-

All traffic

All

All

Anywhe...

0.0.0.0/0

0.0.0.0/0

0.0.0.0/0

Delete

Add rule

⚠️ Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel

Preview changes

Save rules

EC2 > Security Groups > sg-01ab88557582b3289 - launch-wizard-3 > Edit inbound rules

Edit inbound rules

Edit inbound rules

Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Info

Security group rule ID

sgr-08c0b09b0368ac1df

Type

Info

SSH

Protocol

Info

TCP

Port range

Info

22

Source

Info

Custom

Description - optional

Info

Delete

-

All traffic

All

All

Anywh...

0.0.0.0/0

X

Q. 0.0.0.0/0

0.0.0.0/0

X

Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

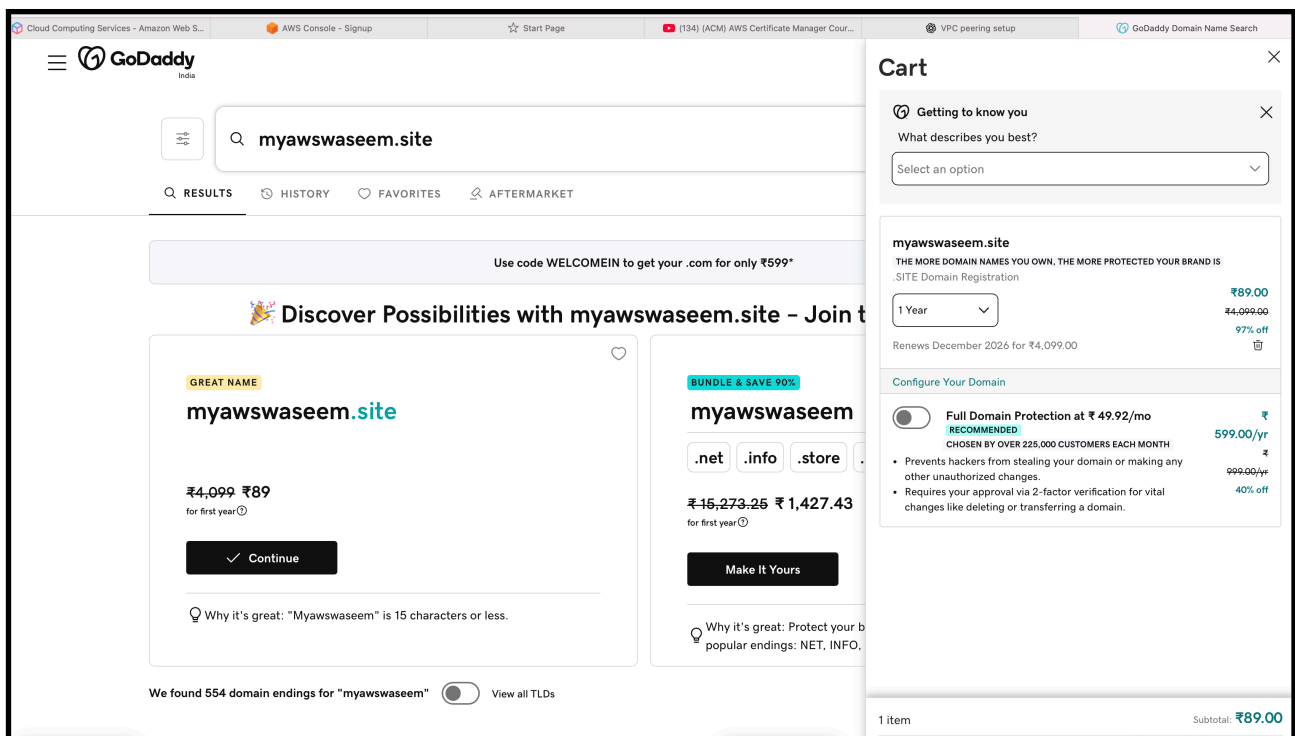
```
waseemakram@waseem ~ % cd downloads
waseemakram@waseem downloads % ssh -i "virginia.pem" ec2-user@ec2-44-202-76-137.compute-1.amazonaws.com
The authenticity of host 'ec2-44-202-76-137.compute-1.amazonaws.com (44.202.76.137)' can't be established.
ED25519 key fingerprint is SHA256:k5aau34mWN0Myn5nfbz1kshW4WBpx/3EmsebzqNrfooo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-44-202-76-137.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
#_
~\ #### Amazon Linux 2023
~~ \#####\
~~ \###|
~~ \#/ ____ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' '->
~~~
~~~. _ . /
~~ / /
~~ /m/'

[ec2-user@ip-172-31-9-43 ~]$ ping 10.0.1.153
PING 10.0.1.153 (10.0.1.153) 56(84) bytes of data.
64 bytes from 10.0.1.153: icmp_seq=1 ttl=127 time=109 ms
64 bytes from 10.0.1.153: icmp_seq=2 ttl=127 time=109 ms
64 bytes from 10.0.1.153: icmp_seq=3 ttl=127 time=109 ms
64 bytes from 10.0.1.153: icmp_seq=4 ttl=127 time=109 ms
64 bytes from 10.0.1.153: icmp_seq=5 ttl=127 time=109 ms
64 bytes from 10.0.1.153: icmp_seq=6 ttl=127 time=109 ms
64 bytes from 10.0.1.153: icmp_seq=7 ttl=127 time=109 ms
64 bytes from 10.0.1.153: icmp_seq=8 ttl=127 time=109 ms
64 bytes from 10.0.1.153: icmp_seq=9 ttl=127 time=109 ms
64 bytes from 10.0.1.153: icmp_seq=10 ttl=127 time=109 ms
64 bytes from 10.0.1.153: icmp_seq=11 ttl=127 time=110 ms
64 bytes from 10.0.1.153: icmp_seq=12 ttl=127 time=109 ms
64 bytes from 10.0.1.153: icmp_seq=13 ttl=127 time=109 ms
64 bytes from 10.0.1.153: icmp_seq=14 ttl=127 time=109 ms
64 bytes from 10.0.1.153: icmp_seq=15 ttl=127 time=109 ms
```

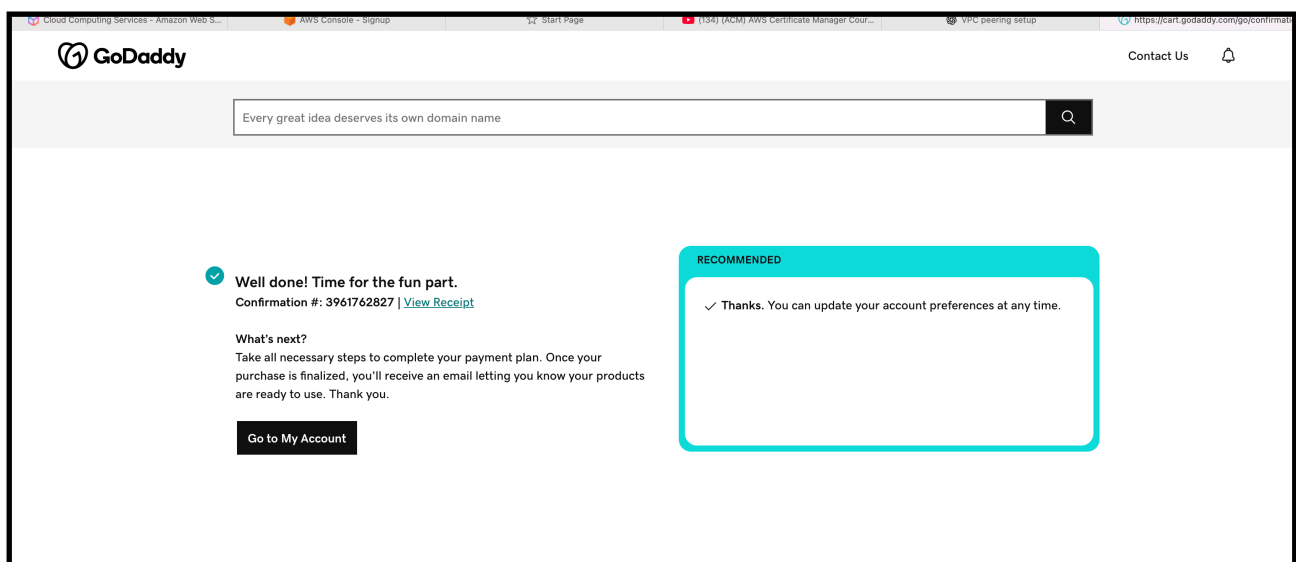
2)Purchase one domain from GoDaddy.

—> go to godaddy website and buy a domain

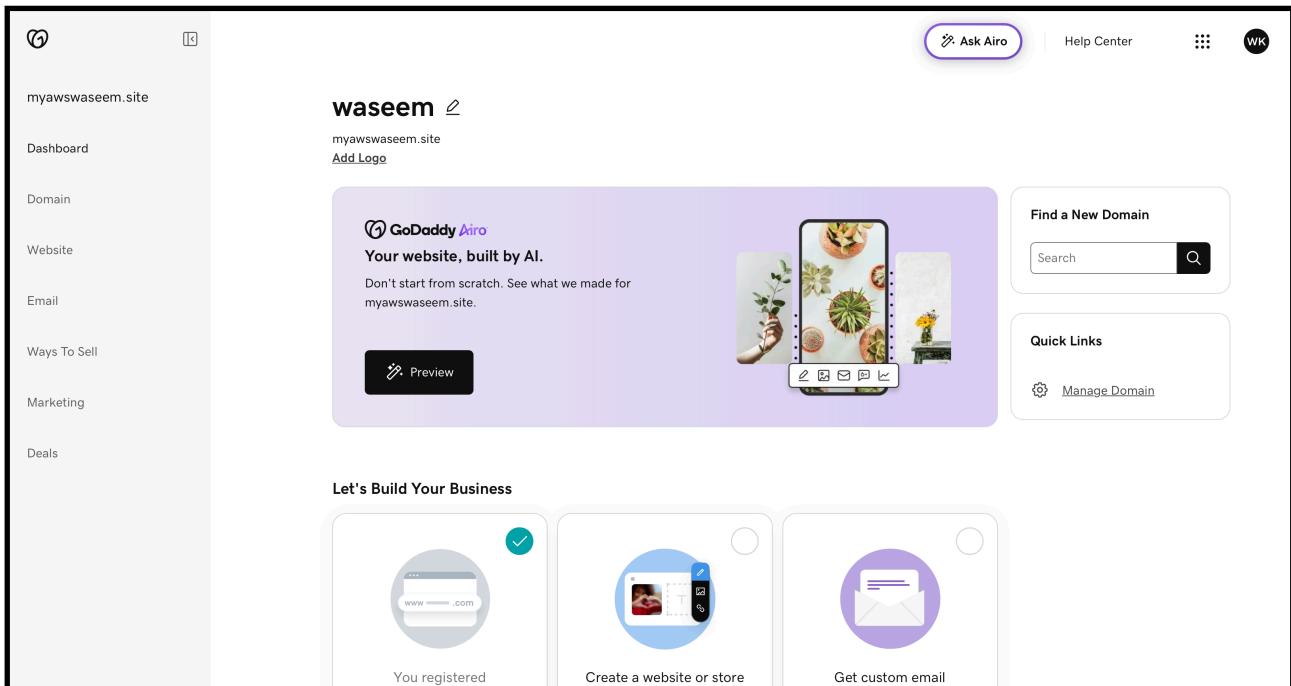
—> my domain name “myawswaseem.site”



—> complete the payment process

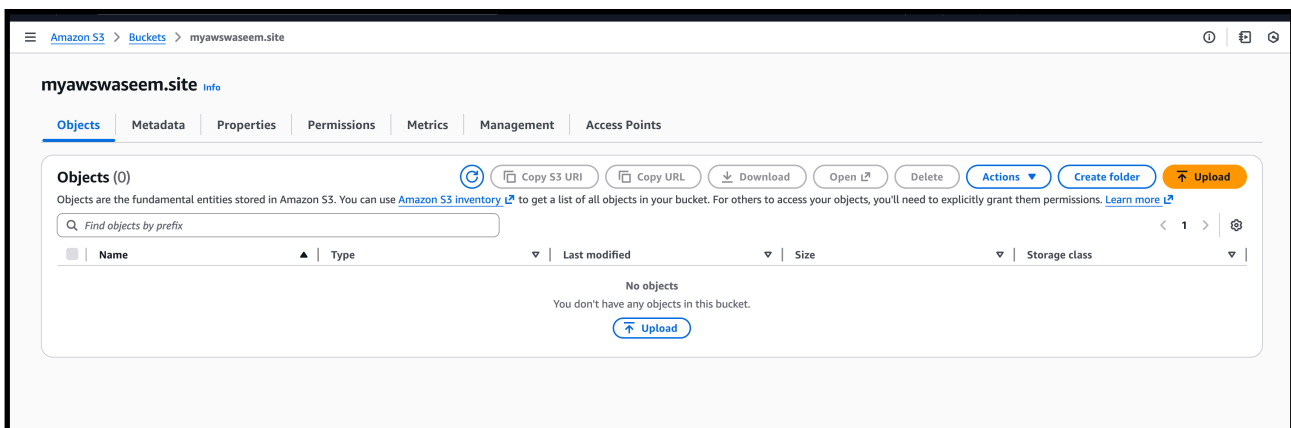


—> purchased one domain successfully



3)Deploy static website in S3.

—> create a s3 bucket name same as domain name which we purchased "myawswaseem.site"



—> now go the static website hosting

Requester pays
When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. [Learn more](#)

Requester pays
Disabled

Edit

Static website hosting
Use this bucket to host a website or redirect requests. [Learn more](#)

We recommend using AWS Amplify Hosting for static website hosting
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about [Amplify Hosting](#) or [View your existing Amplify apps](#)

Create Amplify app

S3 static website hosting
Disabled

—> add index file and error file

Amazon S3 > Buckets > myawsaseem.site > Edit static website hosting

Edit static website hosting

Static website hosting
Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting
☐ Disable
☒ Enable

Hosting type
☒ Host a static website
Use the bucket endpoint as the web address. [Learn more](#)
☐ Redirect requests for an object
Redirect requests to another bucket or domain. [Learn more](#)

For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document
Specify the home or default page of the website.

Error document - optional
This is returned when an error occurs.

Redirection rules - optional
Redirection rules, written in JSON, automatically redirect webpage requests for specific content. [Learn more](#)

1	
---	--

Static website hosting
Use this bucket to host a website or redirect requests. [Learn more](#)

We recommend using AWS Amplify Hosting for static website hosting
Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. Learn more about [Amplify Hosting](#) or [View your existing Amplify apps](#)

Create Amplify app

S3 static website hosting
Enabled

Hosting type
Bucket hosting

Bucket website endpoint
When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)
<http://myawsaseem.site.s3-website.eu-north-1.amazonaws.com>

—> add the same files name from static web site

Upload succeeded
For more information, see the Files and folders table.

Upload: status

Close

After you navigate away from this page, the following information is no longer available.

Summary

Destination
s3://myawswaseem.site

Succeeded
2 files, 39.0 B (100.00%)

Failed
0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (2 total, 39.0 B)

Find by name

Name	Folder	Type	Size	Status	Error
index.html	-	text/html	12.0 B	Succeeded	-
error.html	-	text/html	27.0 B	Succeeded	-

—> now copy the static website url and paste in browser



- 4) Create a CDN and attach one SSL certificate.
- 5) Create a Route 53 hosted zone and map the domain with the CDN.

—> we have to go and create a certificate manager certificate
By our domain name “ myawsaseem.site”

The screenshot shows the 'Request public certificate' page in the AWS Certificate Manager console. The breadcrumb trail is 'AWS Certificate Manager > Certificates > Request certificate > Request public certificate'. The page title is 'Request public certificate'. Under the 'Domain names' section, there is a text input field for 'Fully qualified domain name' containing 'myawsaseem.site' and a 'Remove' button. Below it, another input field contains 'www.myawsaseem.site' with another 'Remove' button. An 'Add another name to this certificate' button is also present. A note states: 'You can add additional names to this certificate. For example, if you're requesting a certificate for "www.example.com", you might want to add the name "example.com" so that customers can reach your site by either name.' The 'Allow export' section has two radio buttons: 'Disable export' (selected) and 'Enable export'. The 'Validation method' section has a radio button for 'DNS validation - recommended' (selected).

- > route53 —> hosted zones
—> create a hosted zone of that domain name

The screenshot shows the 'Create hosted zone' page in the AWS Route 53 console. The breadcrumb trail is 'Route 53 > Hosted zones > Create hosted zone'. The page title is 'Create hosted zone'. Under the 'Hosted zone configuration' section, there is a text input field for 'Domain name' containing 'myawsaseem.site'. Below it, a note states: 'Valid characters: a-z, 0-9, ! * # \$ % & ' () + , - . / : ; < = > ? @ [\] ^ _ ` { | } . ~'. The 'Description - optional' section has a text input field containing 'The hosted zone is used for...'. Below it, a note states: 'The description can have up to 256 characters. 0/256'. The 'Type' section has two radio buttons: 'Public hosted zone' (selected) and 'Private hosted zone'.

aws

Search

[Option+S]

Route 53 > Hosted zones > myawswaseem.site

Route 53

Dashboard

Hosted zones

Health checks

Profiles

Global Resolver

Global resolvers New

VPC Resolver

VPCs

Inbound endpoints

Outbound endpoints

Rules

Query logging

Outposts

Domains

Registered domains

Requests

IP-based routing

CIDR collections

myawswaseem.site was successfully created.

Now you can create records in the hosted zone to specify how you want Route 53 to route traffic for your domain.

Public myawswaseem.site

Delete zone

Test record

Configure query logging

Hosted zone details

Edit hosted zone

Records (2)

Accelerated recovery

DNSSEC signing

Hosted zone tags (0)

Records (2)

Info

Delete record

Import zone file

Create record

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value

Type

Routing p...

Alias

< 1 >

☐

Record name

Type

Routin...

Differ...

Alias

Value/Route traffic to

TTL (s...

☐

myawswaseem.site

NS

Simple

-

No

ns-471.awsdns-58.com.
ns-602.awsdns-11.net.
ns-1537.awsdns-00.co.uk.
ns-1527.awsdns-62.org.

172800

☐

myawswaseem.site

SOA

Simple

-

No

ns-471.awsdns-58.com. awsd...

900

—> now add this route traffic to —> godaddy —> myawswaseem.site
—>dns —> add name servers

myawswaseem.site

Connect Domain

Overview

DNS

Registration Settings

Products

Activity Log

DNS Records

Forwarding

Nameservers

Hostnames

DNSSEC

Nameservers

determine where your DNS is hosted and where you add, edit or delete your DNS records.

Using custom nameservers

Change Nameservers

Nameservers

ns-1658.awsdns-15.co.uk

ns-165.awsdns-20.com

ns-1344.awsdns-40.org

ns-704.awsdns-24.net

—> now go to cloud front and create distribution
—>now

CloudFront > Distributions > Create distribution

Step 1 **Get started**
Step 2 Specify origin
Step 3 Enable security
Step 4 Get TLS certificate
Step 5 Review and create

Get started

Connect your websites, apps, files, video streams, and other content to CloudFront. We optimize the performance, reliability, and security for your web traffic.

Distribution options [Info](#)

Distribution name
Name will be stored as a tag on the resource. You can change the name, or more tags, later.

myawswaseem-site-distribution

Description - optional

Distribution type

☒ **Single website or app**
Choose if each website or application will have a unique configuration.

☐ **Multi-tenant architecture - New**
Choose when you have multiple domains that need to share configurations. This is a common architecture for SaaS providers.

Domain [Info](#)

Route 53 managed domain - optional
Enter a domain that's already registered with Route 53 in your AWS account. CloudFront will provision a TLS certificate for you. If you have a domain from a different DNS provider, skip this step and configure your own.

myawswaseem.site [Check domain](#)

—> so our aws certificate manager —>certificate is pending after
success add the certificate details to the cloudfront

us-east-1.console.aws.amazon.com/acm/home?region=us-east-1#/certificates/list

Relaunch to update

aws Search [Option+S] United States (N. Virginia) Account ID: 2076-6279-1773 waseem akram

AWS Certificate Manager > Certificates

AWS Certificate Manager (ACM)

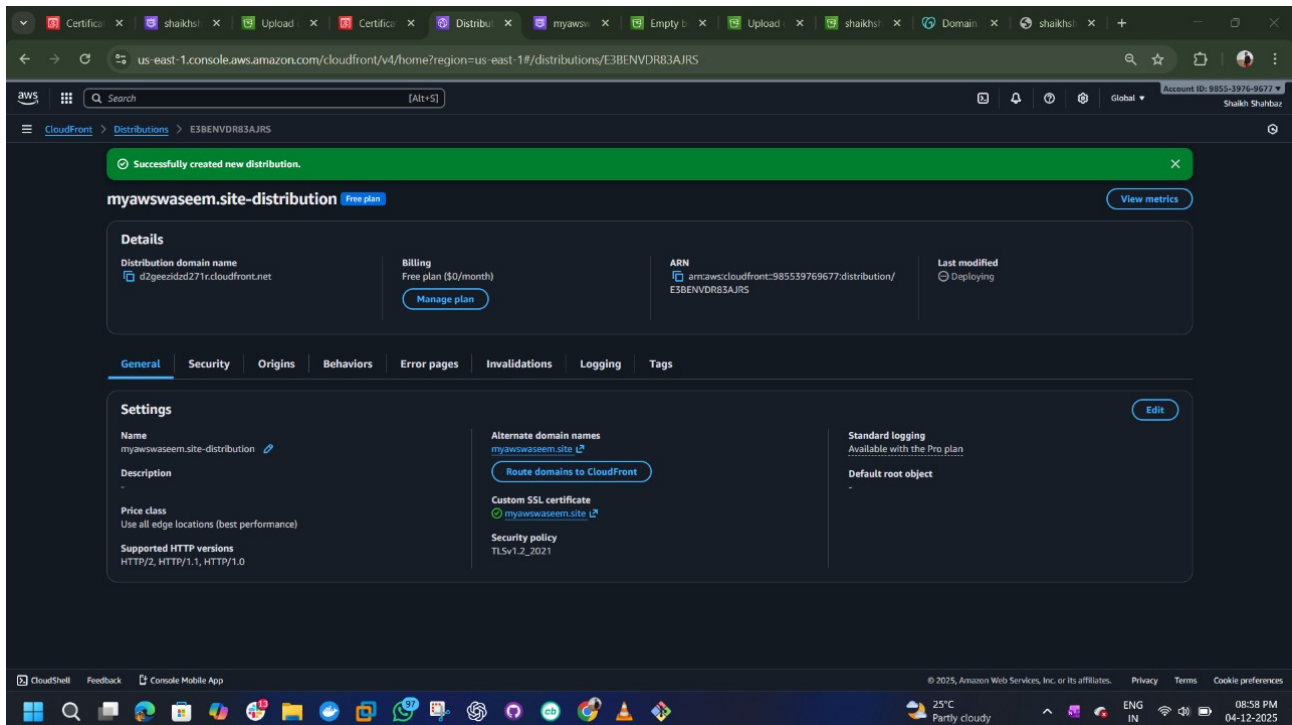
[List certificates](#)
[Request certificate](#)
[Import certificate](#)
[AWS Private CA](#)

Certificates (1)

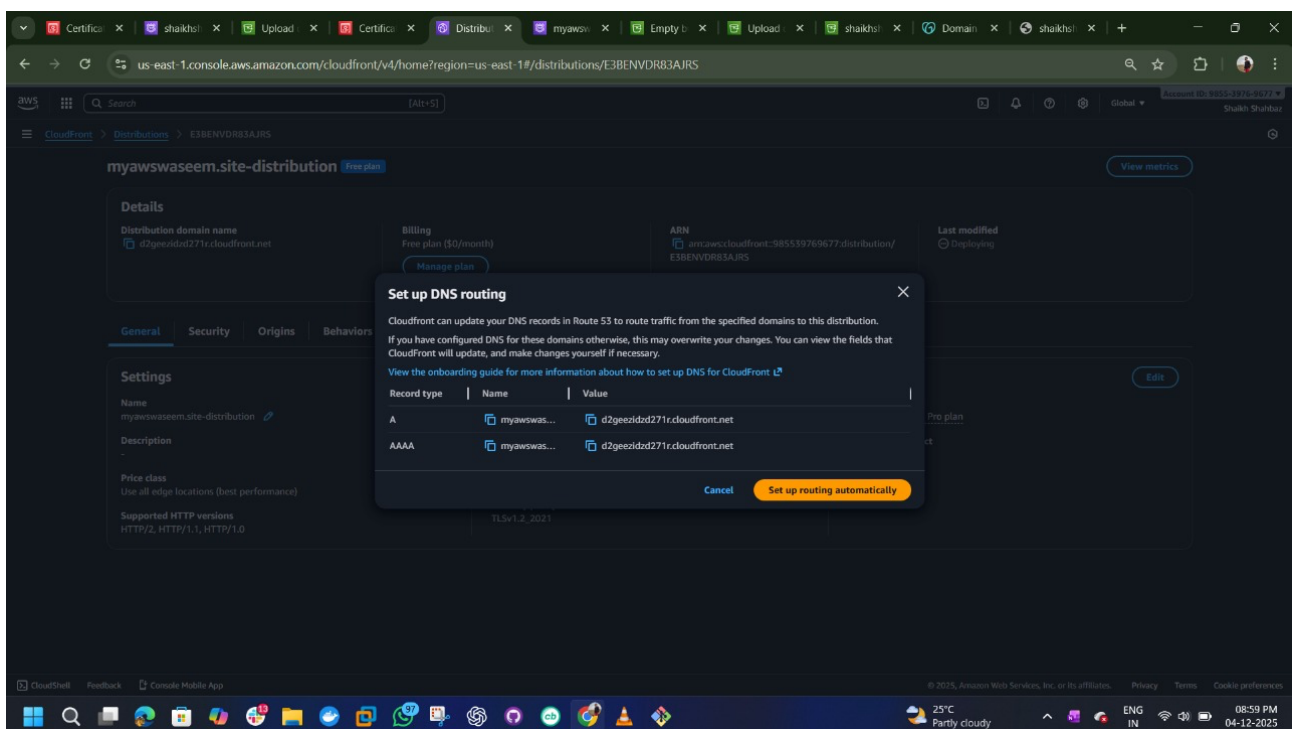
[More actions](#) [Manage expiry events](#) [Import](#) [Request](#)

☐	Certificate ID	Domain name	Type	Status	In use	Renewal eligibility
☐	d3582ad8-0504-46c7-a4b6-40a9e83a588f	myawswaseem.site	Amazon Issued	Pending validation	No	Ineligible

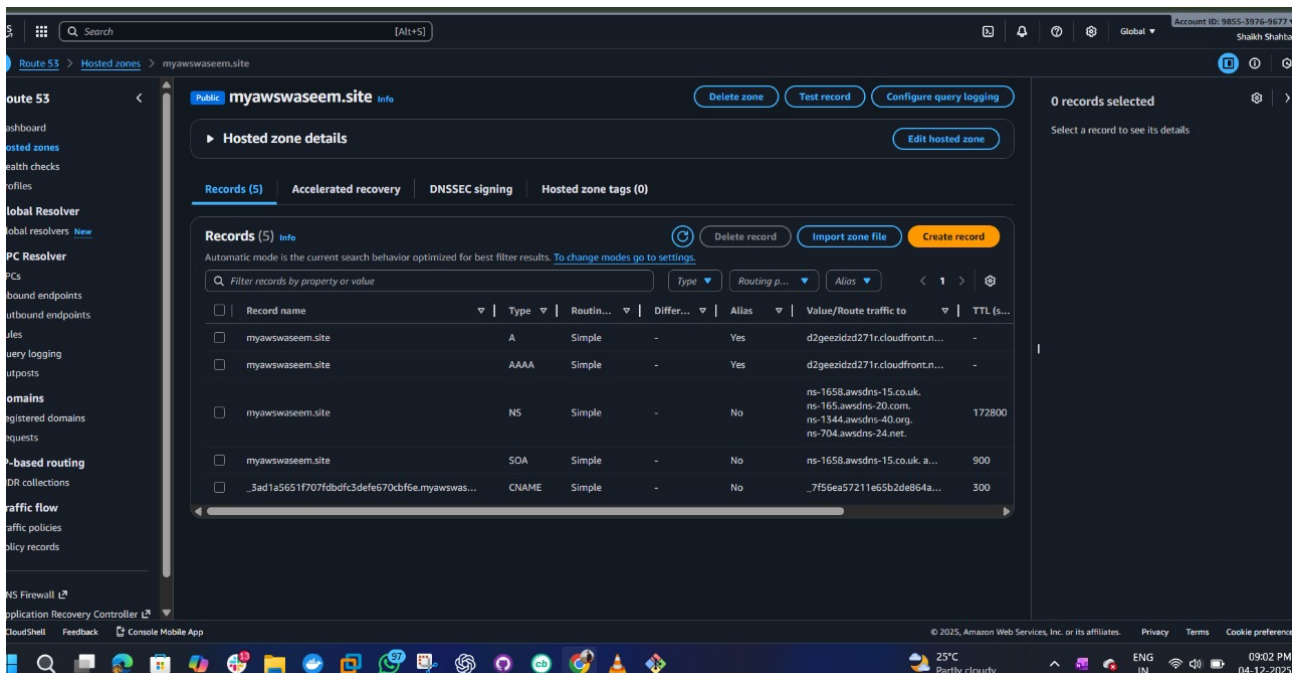
—> after adding the certificate attached



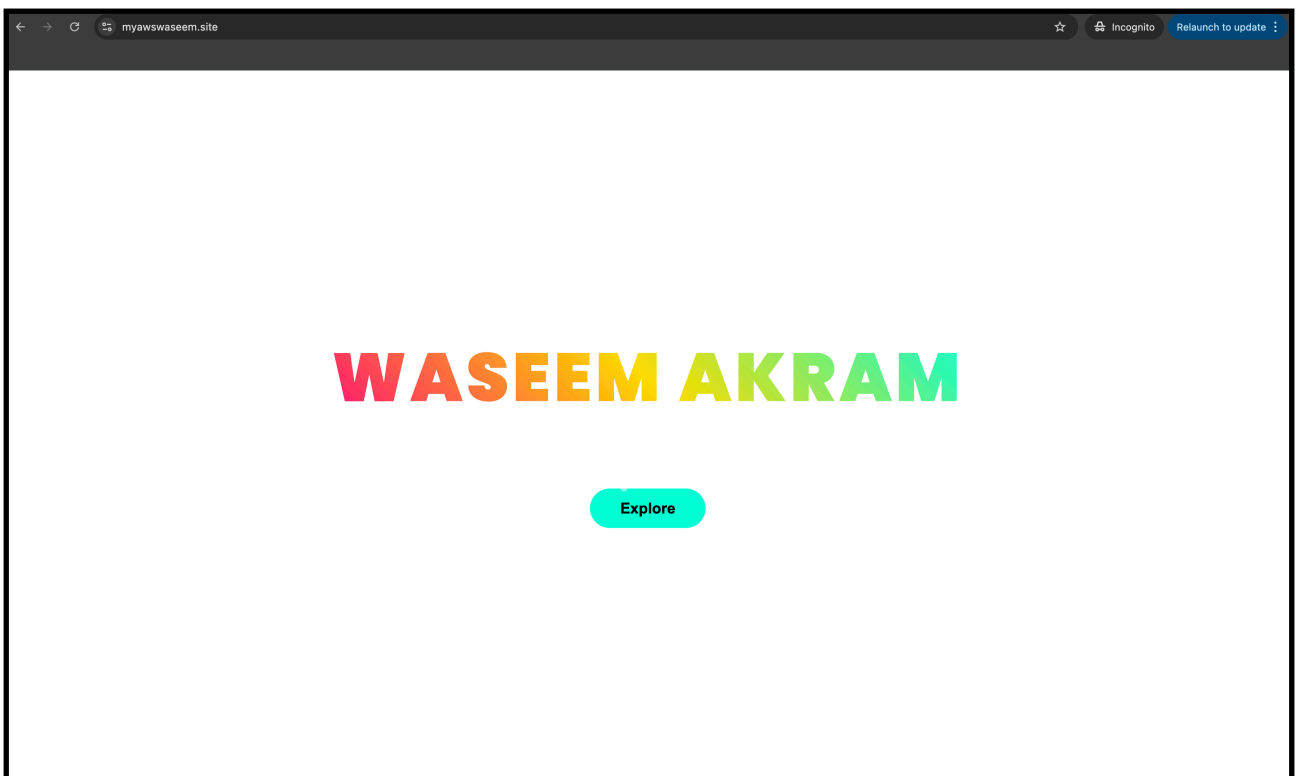
—> give the dns routing to the record type A



—> this our r53 records now we can



6)Update the index.html in the S3 bucket and ensure the updated file is accessible using the domain name.



7)Share the domain name in Slack to test the connectivity
My domain “myawswaseem.site”



waseem akram-16 9:11 PM

@Saddam domain name : myawswaseem.site



Shaikh Shahbazz-16 9:27 PM

@Saddam Domain name : shaikhshahbazz.online



awais jabri-16 9:33 PM

@Saddam domain name: coonline.info